

04b — AC in 15 minutes: phasors, j and angles

Read before 05 §6, 06 §6 and 07 §4 · no new syllabus item: it's the tool those AC parts use · textbook: Hayt ch. 10 · every number checked by script ('verify_ac.py')

Map

1 An AC source is just a cosine

→

2 Keep only size and angle (the phasor)

→

3 j = "turn 90°"

→

4 R, L, C become impedances Z

→

5 Solve like DC

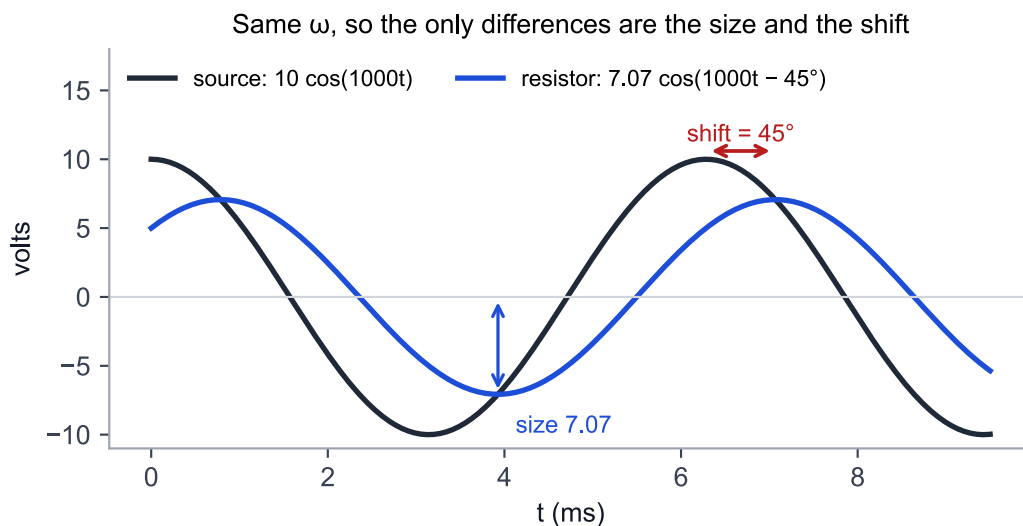
The whole idea in one line: turn the AC circuit into a DC-style circuit with complex numbers, solve it with the rules you already know (Ohm, dividers, KVL, Thévenin), then turn the answer back into a cosine.

1 · An AC source is just a cosine

$$v(t) = V_m \cos(\omega t + \theta)$$

- V_m = size (peak value).
- ω = how fast it wiggles (rad/s). It is given in the question and **never changes** anywhere in the circuit.
- θ = angle: how far the wave is shifted left or right.

In a circuit of R, L and C with one source, **every voltage and current is a cosine at the same ω** . They differ in only two things: **size** and **angle**.



The source and the voltage across the resistor in the example of §5. Same ω ; different size, different angle.

So you only need to track two numbers per quantity. That pair is the **phasor**.

2 · The phasor: write size $\angle$ angle

$$10 \cos(1000t - 45^\circ) \Rightarrow 10 \angle -45^\circ \quad (\text{drop the "cos}(\omega t \dots)\text{", keep size and angle})$$

Going back is the reverse: $7.07\angle -45^\circ$ mA at $\omega = 1000$ means $i(t) = 7.07 \cos(1000t - 45^\circ)$ mA.

SIN INSTEAD OF COS?

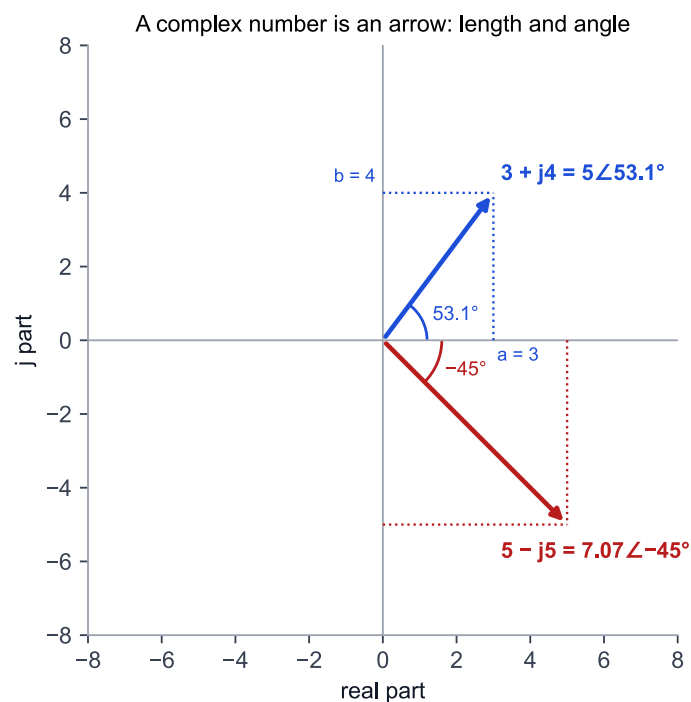
If the question gives sin (like 02's $10 \sin(2000t + 30^\circ)$), just use sin both ways: $10\angle 30^\circ$, and the answer goes back into $\sin(\dots)$. Don't mix sin and cos in one problem.

3 · What j is

A phasor is an **arrow**: length = size, direction = angle. You can write the same arrow two ways:

Form	Looks like	Means	Best for
Polar	$M\angle\theta$	length M, angle θ	multiplying, dividing
Rectangular	$a + jb$	a across, b up	adding, subtracting

j marks the "up" part. That's all it is. (Maths writes i; circuits use j because i is current.)



$3 + j4$: go 3 right, 4 up. The arrow has length 5 and angle 53.1° .

Converting (the only formulas you need):

Rectangular $\rightarrow$ polar: $M = \sqrt{a^2 + b^2}$ $\theta = \tan^{-1}(b/a)$

Polar $\rightarrow$ rectangular: $a = M \cos \theta$ $b = M \sin \theta$

So $3 + j4 = \sqrt{9 + 16}\angle \tan^{-1}(4/3) = 5\angle 53.1^\circ$, and $10\angle 30^\circ = 8.66 + j5$.

QUADRANT TRAP

If a is negative, the arrow points left, and $\tan^{-1}$ gives the wrong angle. Add 180° . Example: $-3 + j4 = 5\angle 126.9^\circ$, not $5\angle -53.1^\circ$. A quick sketch of the arrow catches this.

Two rules for arithmetic:

- **Add/subtract** → **rectangular**. Add the real parts, add the j parts: $(5 - j5) + (5 + j5) = 10$.
- **Multiply/divide** → **polar**. Multiply → **multiply sizes, add angles**. Divide → **divide sizes, subtract angles**.

$$(2\angle 30^\circ)(3\angle 20^\circ) = 6\angle 50^\circ \qquad \frac{10\angle 0^\circ}{5\angle -90^\circ} = 2\angle 90^\circ$$

j itself: $j = 1\angle 90^\circ$, so multiplying by j turns an arrow 90° anticlockwise. Also $j^2 = -1$, and $\frac{1}{j} = -j$ (it turns by -90°). You'll use that last one for capacitors.

CHECK 1 (ANSWERS AT THE END)

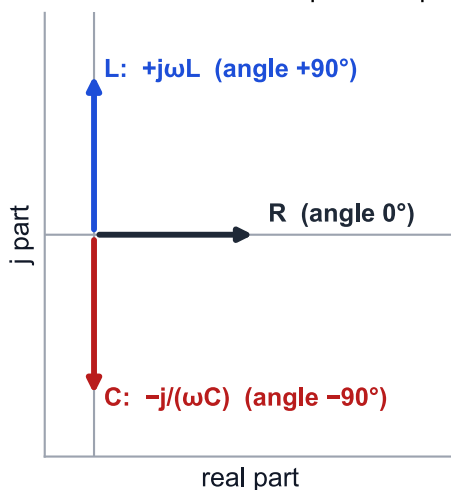
(a) Write $20 \cos(500t + 60^\circ)$ V as a phasor. (b) Convert $6 - j8$ to polar. (c) Divide $10\angle 0^\circ$ by $2\angle 45^\circ$.

4 · R, L and C become impedances

Impedance **Z** is "AC resistance": $V = IZ$, just like $V = IR$. It's complex, because L and C shift the angle.

Element	Impedance Z	Angle	In words
R	R	0°	no shift
L	$j\omega L$	$+90^\circ$	current lags the voltage by 90°
C	$\frac{1}{j\omega C} = -j\frac{1}{\omega C}$	-90°	current leads the voltage by 90°

Where each element's impedance points



R points right, L points up, C points down. Series elements add as arrows.

Example at $\omega = 1000$ rad/s: a 1 H inductor is $j \cdot 1000 \cdot 1 = \mathbf{j1000 \Omega}$. A 1 μ F capacitor is $\frac{1}{j \cdot 1000 \cdot 10^{-6}} = \frac{1000}{j} = -\mathbf{j1000 \Omega}$.

Everything from DC still works with Z in place of R: series Z's add, parallel is $\frac{Z_1 Z_2}{Z_1 + Z_2}$, voltage/current dividers, KVL, KCL, nodal, mesh, superposition, Thévenin. Nothing new to learn: only the numbers are complex.

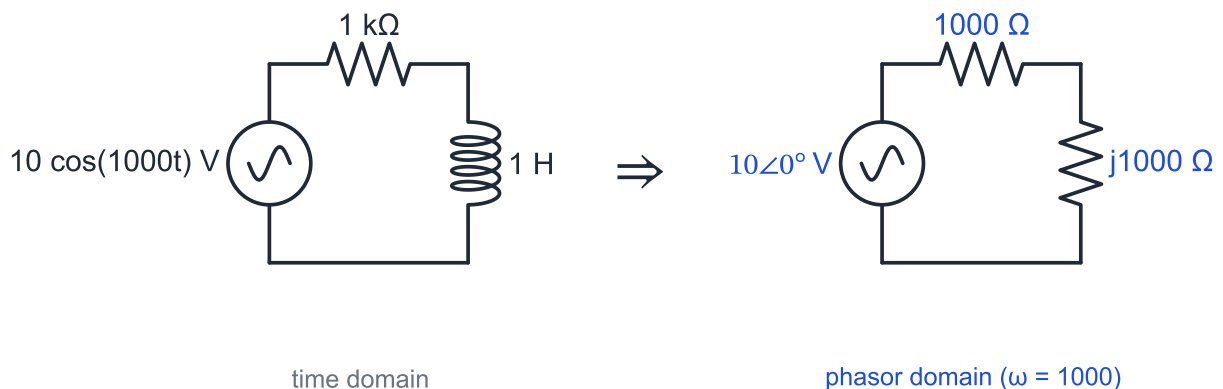
5 • The recipe, on the circuit from 05 §6

QUESTION

10 cos(1000t) V drives 1 k Ω in series with 1 H. Find the current i(t).

GUESS FIRST

Will the current's angle be positive or negative? (Hint: which way does an inductor's Z point?)



Step 1 and 2: the source becomes a phasor, each element becomes its impedance.

Step 1. Source → phasor. 10 cos(1000t) $\Rightarrow$ $10\angle 0^\circ$ V.

Step 2. Elements → Z ($\omega = 1000$). R = 1000 Ω , L: $j\omega L = \mathbf{j1000 \Omega}$.

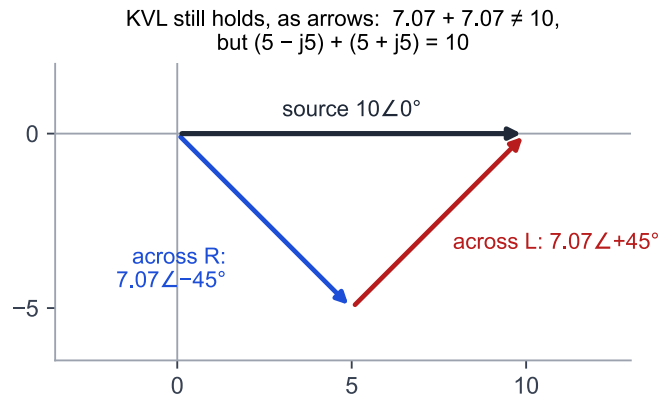
Step 3. Solve like DC. Series, so add (rectangular), then divide (polar):

$$Z = 1000 + j1000 = 1414\angle 45^\circ \Omega \quad I = \frac{V}{Z} = \frac{10\angle 0^\circ}{1414\angle 45^\circ} = 7.07\angle -45^\circ \text{ mA}$$

Step 4. Back to time. i(t) = $7.07 \cos(1000t - 45^\circ)$ mA.

A time simulation of the circuit gives exactly this: peak 7.071 mA, 45° behind the source. The current **lags** (negative angle), because the inductor pulled the angle of Z up to +45°, and dividing subtracts it.

KVL still works, but as arrows. $V_R = I \cdot 1000 = 7.07\angle -45^\circ$ V, and $V_L = I \cdot j1000 = 7.07\angle +45^\circ$ V.



The two element voltages add head-to-tail to the source. Their sizes do not add up to 10.

SIZE TRAP

$7.07 + 7.07 = 14.1$ V, more than the 10 V source! Sizes of phasors don't add. Add in rectangular: $(5 - j5) + (5 + j5) = 10$ ✓.

CHECK 2

Replace the inductor with a **1 μ F capacitor** (same source, same 1 k Ω). Find I as a phasor and $i(t)$. Does the current lead or lag?

6 • Peak vs rms (only for power)

A phasor's size is usually the **peak**. Power formulas use **rms**:

$$\text{rms} = \frac{\text{peak}}{\sqrt{2}} \quad (\text{for a sinusoid})$$

That's why 07 §4 says $P_{\max} = \frac{|V_{\text{Th}}|^2}{4R_{\text{Th}}}$ with rms, or $\frac{|V_{\text{Th}}|^2}{8R_{\text{Th}}}$ with peak.

Summary card

Step	Do
1	Source $V_m \cos(\omega t + \theta) \rightarrow V_m \angle \theta$
2	$R \rightarrow R$, $L \rightarrow j\omega L$, $C \rightarrow -j/(\omega C)$
3	Solve with DC rules. Add in a + jb, multiply/divide in $M \angle \theta$.
4	Answer $M \angle \theta \rightarrow M \cos(\omega t + \theta)$ (same ω)

Where you use it next: 05 §6 (AC superposition), 06 §6 (AC Thévenin/Norton), 07 §4 (conjugate match), mock B11.

Self-test

1. What is the impedance of 0.2 H at $\omega = 500$ rad/s? Of 10 μF at $\omega = 500$ rad/s?
2. $10 \cos(500t)$ V drives 100 Ω in series with 0.2 H. Find $i(t)$.
3. Convert $-3 + j4$ to polar. Why is $\tan^{-1}(4/-3)$ not enough?
4. A current phasor is $5\angle 0^\circ$ mA (peak). What is its rms value?
5. Why can you add phasors only when all sources have the same ω ? (Look at 05 §6.)

Answers

Guess (\$5). Negative: the inductor makes Z's angle positive (+45°), and $I = V/Z$ subtracts it.

Check 1. (a) $20\angle 60^\circ \text{ V}$. (b) $\sqrt{36 + 64} = 10$, $\tan^{-1}(-8/6) = -53.1^\circ \Rightarrow 10\angle -53.1^\circ$ (a is positive, so no correction). (c) $10/2 = 5$, $0^\circ - 45^\circ \Rightarrow 5\angle -45^\circ$.

Check 2. $Z_C = -j1000 \Omega$. $Z = 1000 - j1000 = 1414\angle -45^\circ \Omega$. $I = \frac{10\angle 0^\circ}{1414\angle -45^\circ} = 7.07\angle +45^\circ \text{ mA}$, so $i(t) = 7.07 \cos(1000t + 45^\circ) \text{ mA}$. The current **leads**: a capacitor pulls the angle the opposite way to an inductor.

Self-test.

1. $j\omega L = j \cdot 500 \cdot 0.2 = j100 \Omega$. $\frac{1}{j\omega C} = \frac{1}{j \cdot 500 \cdot 10^{-5}} = \frac{200}{j} = -j200 \Omega$.

2. $Z = 100 + j100 = 141.4\angle 45^\circ \Omega$; $I = \frac{10\angle 0^\circ}{141.4\angle 45^\circ} = 70.7\angle -45^\circ \text{ mA} \Rightarrow i(t) = 70.7 \cos(500t - 45^\circ) \text{ mA}$.

3. $5\angle 126.9^\circ$. $\tan^{-1}(4/-3)$ gives -53.1° , an arrow pointing right and down; but $-3 + j4$ points left and up, so add 180° .

4. $5/\sqrt{2} = 3.54 \text{ mA rms}$.

5. A phasor only means something at one ω ($Z_L = j\omega L$ changes with ω). Different ω 's: solve each separately, turn each answer into a cosine, then add in time.